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Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI(voice user interface)

Command line Interface(CLI)

- A Text-based Interface (Command line interface) enables users to communicate with a computer using the command line, as opposed to using a graphical interface.

- Compact and Powerful – CLI is usually faster and more powerful than a GUI, particularly for expert users who know the correct commands.

- Automation & Scripting – The command-line interface is there to allow users to automate tasks with scripts, making the repetitive tasks easier.

- Lightweight and Resource Friendly – Compared to GUI application, CLI uses very little system resources which is why they are most suited for use on remote servers example: Windows command prompt

IMPLEMENTATION:

cli.py > ...

```python
# Prime Checker CLI

num = int(input("Enter number: "))

if num > 1:
    for i in range(2,num):
        if num % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime Number")
else:
    print("Not Prime")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
PS C:\Users\7muki\Downloads\ex 3> & C:/Users/7muki/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/7muki/Downloads/ex 3/cli.py"
Enter number: 7
Prime Number
PS C:\Users\7muki\Downloads\ex 3>
```

Graphical user interface(GUI)

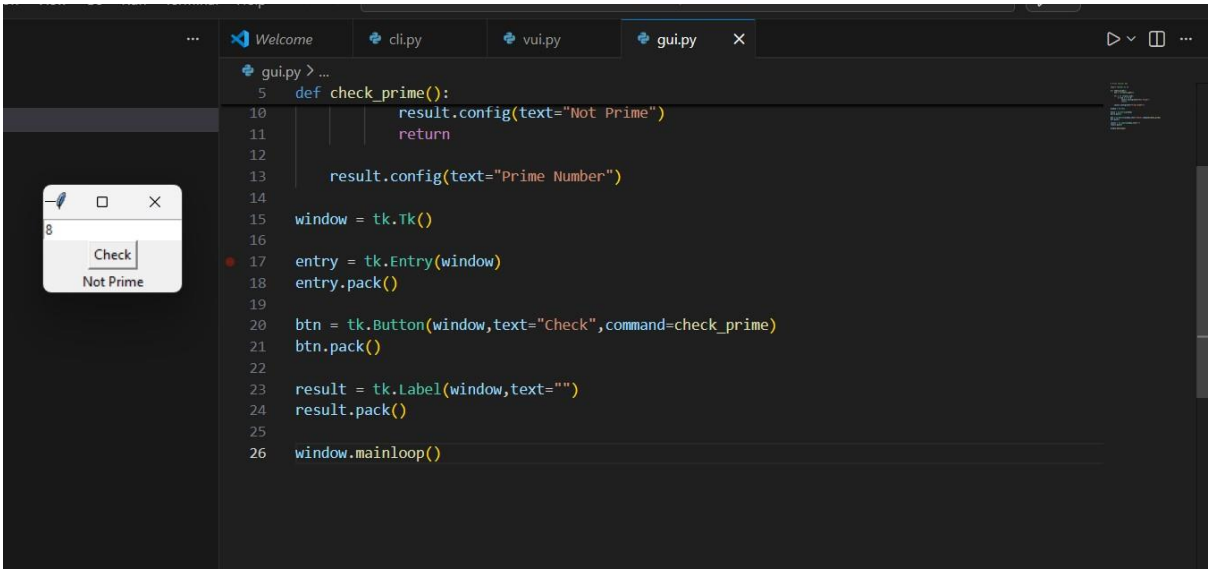
- A GUI uses visual elements including icons, buttons, and windows which makes it more user-friendly for users to work with the system.

- More Resource Intensive – GUI applications use more system resources (CPU, RAM, and GPU) in contrast to a Command Line Interface (CLI) that can degrade performance in low-end devices.

- Limited Automation: GUI does not have automation and scripting abilities like the CLI and is thus more tedious for repetitive jobs.

example: Windows, Android, iOS

IMPLEMENTATION:



Voice user interface(VUI)

- **No Hands – VUI allows me to talk to a system and in some cases this is much more convenient than having to type.**

- **Natural Communication– VUI allows for more natural and fluid communication, as it can process users' speech and verbal commands, eliminating typing or clicking.**

- **Accuracy Challenges – Speech recognition can occasionally have difficulty with accents, background noise, and complex commands, sometimes resulting in misinterpretation.**

example: Amazon, Alexa, Google Assistant

IMPLEMENTATION:

```python
import speech_recognition as sr

# create recognizer object
r = sr.Recognizer()

# take voice input
with sr.Microphone() as source:
    print("Speak a number:")
    r.adjust_for_ambient_noise(source)
    audio = r.listen(source)

# convert voice to text and check prime
try:
    text = r.recognize_google(audio)
    print("You said:", text)

    num = int(text)

    if num <= 1:
        print("Not Prime Number")

    else:
        for i in range(2, num):
            if num % i == 0:
                print("Not Prime Number")
                break
        else:
            print("Prime Number")

except sr.UnknownValueError:
    print("Sorry, could not understand your voice")

except sr.RequestError:
    print("Internet connection error")

except ValueError:
    print("Please speak a valid number")
```

```
Sorry, could not understand your voice
(venv311) PS C:\Users\7muki\Downloads\ex 3> python vui.py
Speak a number:
You said: 2002
Not Prime Number
(venv311) PS C:\Users\7muki\Downloads\ex 3>
```

USER INTERFACE DESIGN EX-3